محافظة القاهرة

وحدة تشغيل وإدارة مدارس التكنولوجيا التطبيقية

مدرسة We للتكنولوجيا التطبيقية

Physics

Model (A)

الإسم :   الفصل: رقم الكشف:

(Put (T) for true and (F) for false)

(1) Fleming's left hand rule used to determine the direction of magnetic force acting on a wire carrying a current and putted in magnetic field.

(2) If the current in the wire is doubled the magnetic force affected on it will halved.

(3) The magnetic flux density unit is equivalent to (N / A . m).

(4) If the magnetic flux increases four times the magnetic dipole moment also increases four times.

(MCQ)

(5) A rectangular loop ($17*10^{-4}$) of 35 turns, carrying current of 2A, is placed in magnetic field of 0.35T flux density such that the plane of the loop is parallel the field the magnetic dipole moment equal …………. A . m^2

(a) Zero **(b) 0.119** **(c) 416.5** **(d) 0.0416**

(6) A straight wire 35 cm long through which passes a current 10A . The wire is placed perpendicular to a magnetic field of magnetic flux density 0.5 T. Then the force acting on the wire

(a)1.75 N **(b) 175 N** **(c) 0.175 N** **(d) 1750 N**

(Put (T) for true and (F) for false)

(7) Galvanometer of the middle zero scale reads the magnitude of Direct current only.

(8) Emf induces as the value of the magnetic flux is great.

(9) Emf induces must be in a direction such as to oppose the change producing it.

(10) Increasing the current intensity in primary coil by using rheostat produces forward induces current.

(11) Switching off the current in primary coil produces forward induces current

In the primary coil.

(12) The scientific base of electric transformer is the self-induction.

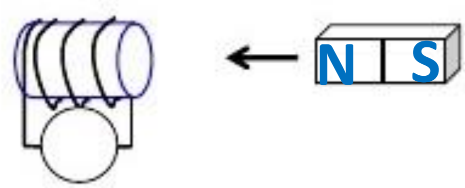
(MCQ)

(13) In faraday Experiment: the Emf induced is direct proportional to

(a) The strength of the magnet **(b) The rate of change of magnetic flux**

(c) The magnetic flux density **(d) the length of the coil**

(14) In the opposite figure: what is the pole formed in the area faced the magnet



a) South pole

b) North pole

d) we can't determine

(15) In faraday Experiment: the direction of induced current which generated in the conductor depends on

(a) The direction of the motion only **(b) The direction of the magnetic field only**

(c) The direction of the magnetic field and the motion **(d) None of the above**

(16) the equivalent unit for (Weber)

(a) Volt. sec **(b) Volt / sec** **(c) Tesla . m** **(d) Tesla / m^2**

(17) An induced EMF of value 6 V is generated in a coil whose number of turns 212 turns , when the rate of change of the magnetic flux through it is

(a) - 0.028 Wb/sec **(b) + 0.028 Wb/sec** **(c) + 0.015 Wb/sec** **(d) - 0.01 Wb/sec**

(18) Backward current is in the opposite direction of

(a) Induces current in primary coil **(b) Original current in primary coil**

(c) Induces current in secondary coil **(d) Original current in secondary coil**

(19) An inductance of 0.06 H exist between two coils if a current of 6 A is decay in 0.012 sec what is value of the EMF induced

(a) 30 volt **(b) 3 volt** **(c) 50 volt** **(d) 12 volt**

(20) Henry is equivalent to

(a) Volt . sec **(b) A . sec** **(c) Volt . Sec / A** **(d) A . Sec / volt**